Student Loans in the United States

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Introduction

College causally increases wages, but not everyone who wants to make this productive investment has access to savings or credit to pay for college.[1] Government is often involved in lending for education because human capital (unlike a house or car) cannot be used as collateral; hence, credit is underprovided by the private market. Loan repayments can impose financial burdens on students and families, with especially bad consequences when borrowers default (including wage garnishment and damaged credit).

In 2024, the total amount of outstanding student debt reached $1.59 trillion, up from under $400 billion in 2005. Not only is the amount of debt large, but it is also widespread, with approximately 38 million people having student debt.[2] Accompanying this increase in borrowing, there has been an increase in delinquency on debt. This rapid increase in the use of student loans has prompted discussion on whether student loans are accomplishing their intended goals—do they help students complete education and gain access to high paying jobs?

Key findings

Key finding #1: *Increases in access to borrowing often improve students' educational, labor market, and financial outcomes.* Borrowing that arises from increases in access to student loans may allow students to focus on schooling, avoid dropout, improve their grades, increase the number of credits earned, transfer to four-year institutions, increase their graduation rates, and even raise their earnings. Increased borrowing reduces reliance on credit cards and reduces the number of hours worked for wages in college, which likely helps explain the positive effects on student outcomes. Importantly, these results apply when borrowing increases occur while holding the cost of college fixed.

Key finding #2: *Increases in student debt that arise from increasing costs of college reduce educational attainment and damage household finances.* When the price of college increases, students may use additional student loans to cover the increase. When students use loans for this purpose, it is best to think of this borrowing as one of the effects of an increase in the cost of college, rather than as one of the effects of increased student debt. Increases in student debt that arise from increases in prices reduce educational attainment and negatively affect household finances.

Key finding #3: *Evidence suggests that increasing access to student loans may increase the price that students pay for college, but the evidence is clearer for private, for-profit, and graduate programs than for undergraduate education at public institutions.* This effect is termed the "Bennett Hypothesis," which states that when loans become more available, colleges increase

their prices. This rise in prices could happen because loans increase a student's willingness to pay for higher education. Recent studies have found evidence that supports this hypothesis, particularly for graduate education and the for-profit sector. The practical magnitude of such effects for undergraduate education at public institutions is less clear.

Key finding #4: *Compared to fixed repayment schedules, income-driven repayment (IDR) improves borrowers' financial health.* IDR changes borrowers' monthly payment to a fraction of their income, rather than a fixed amount. This repayment scheme is designed to act as insurance against income shocks that borrowers may experience. A series of papers has shown that enrollment in IDR plans, relative to the default fixed payment plan, reduces delinquency on debt, which is consistent with the intention of these policies. Additionally, there is some evidence that borrowers see improvements in their financial outcomes in the longer run.

Key finding #5: *The presentation of information about loan options has large impacts on loan use.* The way that borrowing options are presented affects the amount that students borrow. The most consistent finding is that whatever option is presented as the default has an outsize influence on students' loan borrowing and repayment decisions.

Policy background

The U.S. federal government has several loan programs. Undergraduate borrowers can borrow through two primary programs, Direct Subsidized Loans and Direct Unsubsidized Loans. Subsidized loans are available to students who demonstrate financial need, and they currently have the same interest rate as unsubsidized loans. Subsidized loans do not accrue interest while the student is enrolled in college or for six months after enrollment ends.

Financial need does not determine the total amount that a student can borrow (because unsubsidized loans are available regardless of need); rather, this amount depends on a student's status as dependent (traditional student) versus independent (nontraditional student) and the student's year in college. Furthermore, federal loans have yearly and lifetime limits for undergraduates. Dependent first-year students can borrow up to $5,500, with no more than $3,500 being subsidized. Dependent second-year students can borrow up to $6,500, with no more than $4,500 being subsidized. Dependent third-year students and beyond can borrow up to $7,500, with no more than $5,500 being subsidized. In total, dependent undergraduates can borrow only $31,000 in direct loans in their lifetime. Independent undergraduates have higher yearly and lifetime borrowing amounts.

Graduate students can borrow using direct loans (with a yearly limit of $20,500 and a lifetime limit of $138,500 with higher limits for some health programs). Additionally, graduate students can borrow up to the Cost of Attendance of their program using the Grad PLUS loan program with no lifetime maximum on borrowing.

Finally, parents can borrow when their children attend college using Parent PLUS loans. Parents can borrow if they do not have an adverse credit history, and they can borrow up to a student's cost of attendance less any other financial assistance that they receive. Parents who borrow for their children have a different motivation for borrowing. These parents are not investing in their own human capital, with the expectation that increases in their earnings can pay back any debt accumulated; rather, they may be more motivated by transferring income to future generations.

Federal loans account for 85% of college borrowing.[3] Delinquency and default vary by student characteristics such as the institution at which the student is enrolled, with less selective institutions generally having higher default rates.[4] Borrowers with small balances are more likely to default.[5] Borrowers with very high balances do not default as often and accumulate much of their debt in graduate school.[6] Notably, default rates are higher for Black borrowers than for White borrowers.[7] Institutions have discretion in how they "package" loans to students. Colleges can decide how much, if any, student loans appear on a student's financial aid award letter.

Students can repay federal student loans using a variety of plans. The default is a traditional plan that requires monthly payments of a fixed size over a time frame (the default is 10 years). Other repayment plans require a fraction of income (above a minimum income threshold). These differences in repayment plans offer different amounts of insurance against negative income shocks. The fraction of borrowers using income-based repayment plans grew for undergraduates from 11% to 24% from 2010 to 2017.[8]

Evidence on the key findings

Key finding #1: Increases in access to borrowing often improve students' educational, labor market, and financial outcomes.

Studying the effect of access to student loans on student enrollment, graduation, and earnings has been difficult because nearly all students are eligible for federal student loans. Hence, it is hard to find a good comparison group. Currently, there is very little evidence in the U.S. context on how much the availability of student loans affects the decision to *attend* college. We have more evidence regarding how loan availability affects students after they have made the decision to start college. Access to additional loans may help students complete college by allowing them to focus on their studies and/or reduce the number of hours spent working for wages during college.

One group of researchers studied the effect of increases in the annual maximum amount students could borrow that occurred in 2005–06 and 2007–08.[9] They compared students who borrowed the maximum amount in their first year to students who borrowed less than the maximum amount before and after the changes. This strategy inherently reveals only the effect of higher borrowing limits on students who were already enrolled in college. The authors divided their analysis into four groups defined by dependent versus independent students and students who initially entered a two-year institution versus a four-year institution. They found that dependent students who initially entered a four-year institution borrowed more, were more likely to

graduate from college, and had higher post-college earnings. Increased borrowing limits reduced earnings while in college. This finding suggests that student loans may allow students to focus on schooling. Dependents entering a two-year institution also increased their borrowing, and the authors saw some evidence of improved educational outcomes and earnings for these students, but the results were less precisely estimated. In contrast, independent students with higher annual maxima did not increase their borrowing. The authors also found decreases in delinquency on debt using a national sample of borrowers and no effects of increased access to student loans on mortgage debt.

Another researcher studied an increase in financial aid for students at four-year institutions using the age cutoff for independent students and a regression discontinuity design.[10] For the poorest students in the sample (who previously had an expected family contribution of 0), there was no increase in grant aid but an increase in borrowing. This increase in borrowing led students to graduate earlier than they otherwise would have. In a subsequent study, the authors used the number of credits completed and a regression discontinuity design to compare students who were eligible for additional loans (e.g., freshmen versus sophomores) at public universities in Utah.[11] They found no effect of increased access to student loans on students' grade point average (GPA), credits, persistence or graduation. The change in borrowing was much smaller in this study than in the study presented in the previous paragraph, which may explain some of the discrepancy.[12] An earlier study is notable for examining the role of access to private student loans, which may have effects that are different from those of federal student loans.[13] The authors examined banking deregulation and found that it caused an increase in both private student loan borrowing and enrollment.

In the two-year sector, there has been more work on the effects of student loans. The reason is that some institutions opt out of offering student loans, while others vary the way student loan offers are presented to students. Institutions often make this choice for fear that high default rates will make them ineligible to disburse Title IV aid, which includes Pell Grants and federal student loans. In recent years, very few institutions have been subject to sanctions that would affect their ability to offer Title IV aid. However, many institutions either completely opt out of the student loan program or list an amount that is less than the maximum amount that a student is eligible to borrow on the student's financial aid letter.

One study examined what happens when a two-year institution opts out of offering federal student loans, using the variation in school-level opt-outs across institutions and over time.[14] The author found that opting out decreased loans and that students who borrowed attempted more credits overall and attempted and completed math and science courses. Similarly, another study found that high-need students and female students at community colleges that do not offer student loans increased their total months of enrollment and that dependent students increased their bachelor's degree attainment compared with colleges that offer loans.[15]

Two other papers have studied community colleges and induced changes in borrowing via an intervention that changed the way the college presented student loans. In the first, the authors randomly altered financial aid award letters so that some students saw the full amount that they were eligible for, while others saw $0.[16] This intervention did not change the amount that students could borrow, but it induced differences in borrowing and, as a result, allowed the researchers to estimate the effect of additional student loans on attainment. The additional borrowing increased credits attempted, credits earned, student GPAs, and transfers to four-year colleges. In the second paper, the authors partnered with the Community College of Baltimore County and randomly sent messages to students prompting them to make active choices about their borrowing.[17] These messages led students to borrow less, have worse academic outcomes, and default on student loans at higher rates. Interestingly, both papers used different "nudges" to change borrowing but reached similar conclusions about the benefits of increased borrowing.[18]

To summarize the evidence discussed above, the effect of offering students access to loans has been to generally increase attainment, with one recent study of post-college effects finding reductions in delinquencies and increases in earnings. The evidence for the effect of loan access on attainment is strongest in the community college sector. Some evidence suggests that in the absence of access to student loans, students may turn to other ways to finance their education that reduce academic success. There is some evidence for this with increased availability of student loans reducing the use of credit cards and working during college. Furthermore, some of the effect likely comes from increased persistence early on in a student's academic career (or via increased transfers for two-year students). Despite the generally positive findings, however, the evidence is not completely uniform. One study found no effects of increased loan access at four-year public universities in Utah, while another study found no effects for independent students.[19]

Key finding #2: Increases in student debt that arise from increasing costs of college reduce educational attainment and damage household finances.

Increases in student debt could arise for several reasons. Consider two scenarios. First, tuition increases cause some students to borrow more. Second, the price of college is unchanged, but the maximum amount that students can borrow increases, causing some students to borrow more. In both of these cases, student debt increases. However, in the first case, students are paying more for college and in the second case, they are not. This distinction is important for thinking about policy. The prior section discusses the latter case—what happens when students borrow more, holding price fixed. I now discuss what happens when prices change (via grants or tuition) and how this change affects students.

In one paper, the authors used credit data merged with enrollment and graduation data to study the effects of tuition increases.[20] They used shocks to tuition that affected some student cohorts and not others and found that increases in tuition increased borrowing. They then showed that tuition increases reduced graduation rates and homeownership and increased credit card delinquency.[21] Another group of researchers used similar data and instrumented for the tuition

that a student experienced with the tuition of the student's home state.[22] They found that increases in tuition resulted in increases in student loans. They also found a reduction in the homeownership rate. These studies ask a question that is different from the effects of access to student loans, but they still provide insight into how students use student debt to insulate themselves against price shocks. This places them in a category similar to several other studies on changes in prices, which generally find that reducing the price of college improves student outcomes.[23]

The findings of studies that consider increases in the price of college that increase student debt are notably quite different from those of studies that consider the effect of access to additional student loans. When students have to pay more for college, their educational and household financial outcomes generally worsen. When students are given access to additional loans, holding price fixed, their educational, household financial, and earnings outcomes generally improve.

Key finding #3: Evidence suggests that increasing access to student loans may increase the price that students pay for college, but the evidence is clearer for private, for-profit, and graduate programs than for undergraduate education at public institutions.

Increasing the supply of student loans may increase the price of college, which has become known as the "Bennett Hypothesis," after former Secretary of Education William Bennett. There is increasing evidence for this effect, at least in some sectors. One broad-based study examined increases in loan limits for undergraduates at institutions with more versus fewer students already borrowing at the maximum.[24] The authors found that increases in subsidized loan borrowing increased tuition and that increases in unsubsidized loans also increased tuition but less so than increases in subsidized loans. A different paper examined whether the creation of Grad PLUS loans (which increased the amount that students could borrow for graduate school) led to increases in tuition.[25] Using the variation in which programs were more exposed, they found increases in net prices of approximately 60%. Other research studying Grad PLUS has not found evidence of the "Bennett Hypothesis."[26]These studies used undergraduates as a comparison group. However, undergraduate borrowing saw significant changes during this same time and may have also been affected by changes in graduate pricing, raising potential issues with the use of undergraduates as a comparison group.

While the evidence for the "Bennett Hypothesis" is somewhat limited when loans alone are considered, there is additional evidence that prices increase when aid increases (again, especially at private and for-profit institutions). In this regard, evidence is provided by a group of researchers who also consider increases in Pell Grant aid and by another author who finds that some institutions reduce their own financial aid for students who receive a Pell Grant.[27]Other examples include the Post-9/11 GI Bill and eligibility for all Title IV aid.[28]

Key finding #4: Compared to fixed repayment schedules, IDR improves borrowers' financial health.

One researcher used data from a loan servicer and the insight that some call center agents are more effective in inducing borrowers to enroll in IDR.[29] He found that the use of IDR reduced delinquencies, as well as suggestive evidence of increases in credit scores and mortgage holding rates. Other researchers studied a setting where Navient randomly sent prepopulated IDR applications to borrowers who had applied for IDR.[30] This treatment meaningfully increased IDR enrollment. These researchers then showed that IDR enrollment reduced monthly payments and delinquency on debt more broadly, and also increased credit card balances.

In one paper, the authors used U.S. Department of Education data and a policy discontinuity that meant that very similar borrowers had either a zero payment or a positive payment amount.[31] They showed that qualifying for a minimum payment of zero for IDR plans reduced delinquency and default in the short run but increased default in the longer run. This finding suggests that administrative issues are important in determining repayment.

Repayment rates have changed substantially in recent years. A group of researchers showed that from 2004 to 2014, borrowers (not only those on IDR) repaid a smaller fraction of their loans five years after entering repayment.[32] This finding suggests that future policy attention and research will likely focus on repayment plans.

Key finding #5: The presentation of information about loan options has large impacts on loan use.

Choosing how much to borrow is a complex decision that requires forecasting expected costs and assessing future income, among other matters. Given this complexity, there are several studies that have tried to understand how borrowers make this choice. One study showed that packaging loans—the practice of including loans in financial aid offer letters, rather than waiting for students to request them specifically—increased student loan take up.[33] In another study, the authors randomly prepopulated acceptance forms about financial aid to either decline, accept, or make an active choice.[34] Prepopulating a decline reduced borrowing by around 5 percent. One researcher shows that framing a conditional aid program as a loan had meaningful impacts on take up at NYU's law school.[35] In one framing, the financial aid was a loan that would be forgiven if a student worked in public interest law. In another, the aid was framed as a reduction in tuition that would have to be repaid if the student did not work in public interest law. Although financially equivalent, the tuition reduction meaningfully increased public interest law job placement and affected enrollment in the program. Some researchers sent emails to students at a community college informing them that they did not have to borrow the maximum amount, which had no effect on borrowing decisions.[36] Furthermore, emails that referenced amounts borrowed by recent graduates reduced borrowing. The authors concluded that students making choices about borrowing may face cognitive overload. Other researchers sent text messages to

community college students to encourage active borrowing decisions, which led to reduced borrowing.[37]

Similarly, a group of researchers used a lab experiment to show that the choice of repayment plans does not follow standard models of economic decision making.[38] Rather, more behavioral models of choice describe the choices made by participants. Another group of researchers conducted an experiment to study the choice of IDR plans and found that default options are quite important, whereas information on the distribution of earnings for college graduates has no effect.[39]

The primary conclusion from these studies is that the choice architecture of the borrowing and repayment decision has a meaningful influence on this decision. The most consistent finding is that the default option has an outsize influence on the choices ultimately made by borrowers.

Emerging evidence related to recent policy developments: Payment pauses and loan forgiveness

The literature above indicates that student loan repayment structures can affect outcomes. Recently, federal student loan payments were paused for an extended period due to the COVID-19 pandemic, and there have been calls for forgiveness of student loans and attempts by the Biden administration to carry out broad forgiveness. Only a few recent studies provide insight into the possible effects of these policies. One group of researchers studied a situation where debt was discharged because a private student loan lender lost lawsuits that required student debt to be forgiven, as this lender was unable to establish ownership of the loans.[40] This debt forgiveness reduced delinquency on other accounts and increased geographic mobility and income. Another group of researchers studied how a pause in payments for many borrowers affected credit outcomes by using the fact that the pause applied only to loans owned by the federal government.[41] They found that pausing payments increased borrowing for other types of loans, rather than reducing delinquencies on other types of loans. Rigorous research on the causal effects of payment pauses and broad-based forgiveness is currently limited but remains an important area for additional research.

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